Design and Evaluation of an Educational Game to Teach File System Management

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Outline

Motivations

Introduction

- Computers are very important.
    - Used for work, communication, and organizing documents.
- True not only for scientists and laypeople.
- Worth looking into these skills.

Computer and Digital Literacy

Computer Literacy - Skills to use the actual computer [1][2].

- Make and manage files.
- Use emails.
- Navigate the internet.

Digital Literacy - Skills needed to find, evaluate, and share information from digital sources [3][4].

- Participating in social networks.
- Finding digital information.
- Making digital information.

- Both are very important.
  - Online learning [1][5][6].
  - Work [7].
  - Identify misinformation [8].
- Large overlap.

Deficiency

Despite its importance, deficiencies in computer literacy seem to exist.

- Students lack confidence [5].
  - Confident students tend to overestimate actual skill [9].
- Some jobs also show lack of confidence, such as nursing [10].
  - 28.6% reported formal computer training [11].
- With reports that students learn skills in school [1], suggests maybe more teaching needs to happen.

Strategy

- Game-based learning (GBL) can be effective [12][13][14].
  - Has been used to teach UNIX [15].
- Adaptive learning [16].
  - Positively affects attitudes [17][18].
  - Con: Redundant content [16].
- Automatic item generation (AIG) [19].
  - Lessen adaptive learning burden [20][21].
  - Works well with computerized systems [22].

Target

- Serving as a proof-of-concept, so should be simple.
- Digital file system:
  - Useful.
  - Simple to understand.
  - Integrates easily into a game.

Research Questions

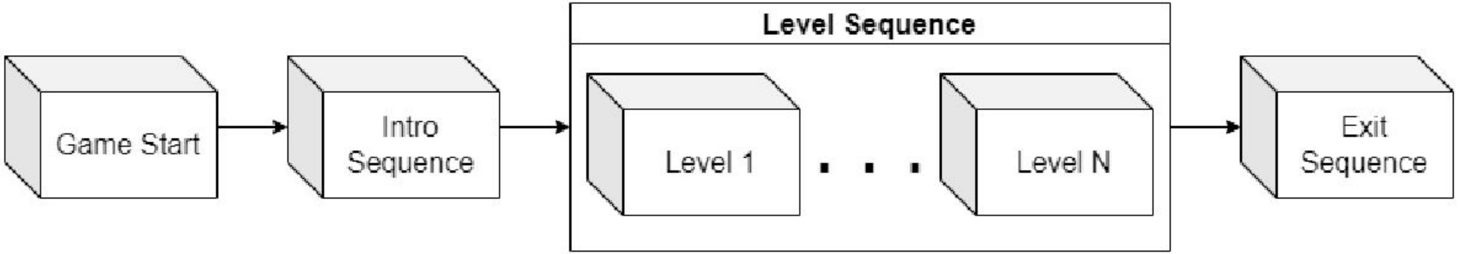
1. How can an educational game be made to procedurally generate puzzles about the digital file system?
2. How could this game determine what level should be generated, given the outcome of the previous level?
3. How does the proposed game improve student understanding of the digital file system?

Research Methods

Concept

- Create a 2D platformer for simplicity.
  - Short, discrete levels to work with AIG.
- Each level involves a file system related puzzle.
  - Must be completed to finish the level.
  - Require action in the real file system.
  - Main challenge.
  - Example: boulder.

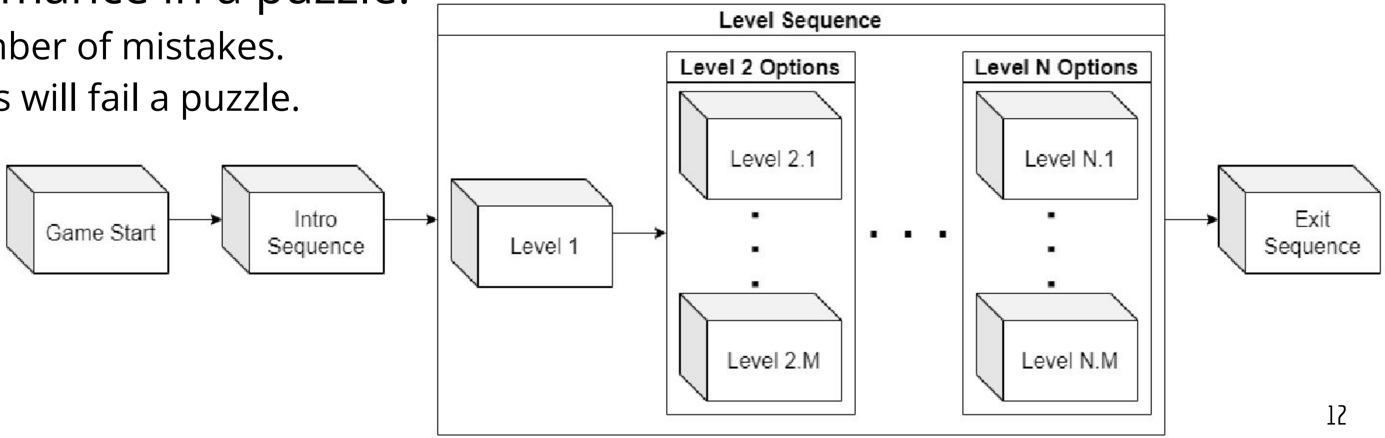
Overall flow of the game.



Adaptive Difficulty

- Puzzle difficulty should adapt to the player.
- Variable traits:
  - Depth (≤ 3)
  - Unknown path (≤ Depth)
  - Puzzle type (delete, create, or move)
- Measure performance in a puzzle.
  - Base it on number of mistakes.
  - Three mistakes will fail a puzzle.

Game flow after the introduction of adaptive difficulty.



Difficulty Calculation

$$\text{Difficulty} = \text{type\_wt} + (0.5 * \mathit{Depth}) + (1.5 * \mathit{Unknown})$$

$$\text{type\_wt} = \begin{cases} 1, & \text{if } T = \text{DELETE} \\ 4, & \text{if } T = \text{CREATE} \\ 8, & \text{otherwise} \end{cases}$$

Example: Creation puzzle 3 folders deep where 1 part is unknown.

Difficulty = 4 + (0.5*3) + (1.5 * 1) = 4 + 1.5 + 1.5 = 7

Calculation of Next Difficulty

$$\text{Target Difficulty} = \begin{cases} D + 0.2D * (3 - \text{Mistakes}), & \text{if Mistakes} < 3 \\ D - 0.2D, & \text{otherwise} \end{cases}$$

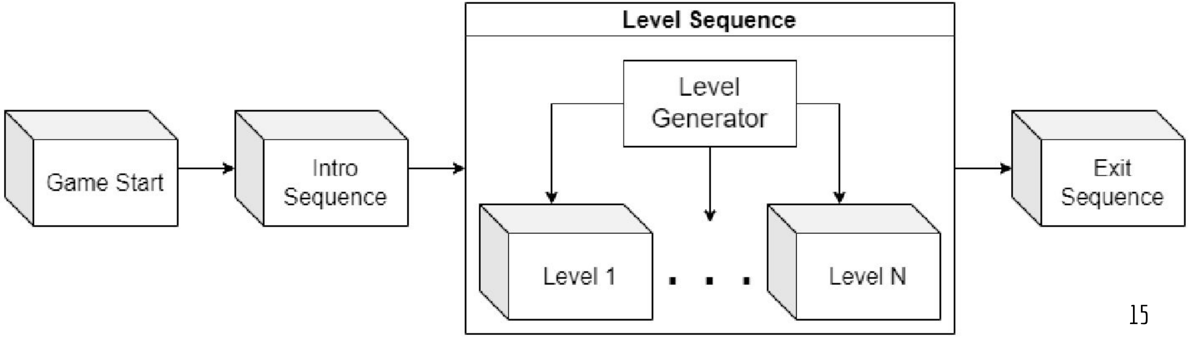
Example: Student completes a difficulty 7 puzzle with 2 mistakes.

Target Difficulty = 7 + 0.2 * 7 * (3 - 2) = 7 + 1.4 * 1 = 8.4

Automatic Item Generation

- Define individual levels to be items.
- Two parts:
  - Puzzle
  - Playspace
- Puzzle varies depth, unknown path, and puzzle type to tweak difficulty.
- Playspace changed for variety.
  - Must work with the puzzle.

Game flow after the introduction of automatic item generation.



System

Overview

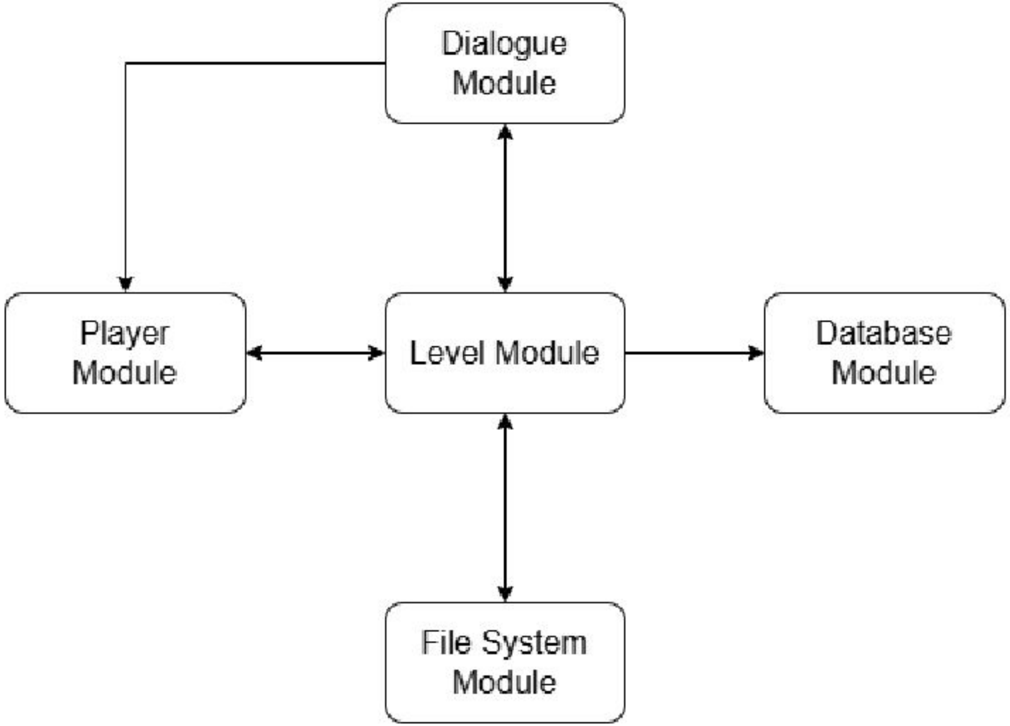
- Developed with the Unity game engine.
    - Due to this, code was written in C#.
- Main focus is the level sequence.
    - Starts with minimum difficulty (1).
    - Difficulty changes based on performance.
    - Termination conditions:
        - 10 levels completed.
        - Maximum difficulty (14) beaten

Minimum Difficulty Structure

System Architecture

- Five modules that make up the system.
  - Player module - handles player inputs.
  - Level module - generates and manages levels.
  - Dialogue module - displays dialogue.
  - File system module - handles file system interactions.
  - Database module - handles the experiment database.
- Note that there are Unity components outside of this, but they are not the focus of this research.

Intermodule Connectivity



Level Generation

First time:

1. Handle level count.
2. Level module makes a puzzle of difficulty 1.
3. File system module helps pick a file and sets up the structure.
4. Dialogue module makes the introduction.
5. Level module picks a level to go alongside the puzzle.

Other times:

1. Handle level count.
2. Previous puzzle generates the new one.
3. File system module helps pick a file and sets up the structure.
4. Dialogue module makes the introduction.
5. Level module picks a level to go alongside the puzzle.

Level Generation Pseudocode

```
GEN_NEXT_PUZZLE()
if levelCount ⩾ maxLevels then
    TriggerEndSequence()
end if
levelCount ← levelCount + 1

currentPuzzle ← currentPuzzle.GEN_NEXT_PUZZLE(mistakes)

for f in deleteFiles do
    f.SET_ACTIVE(false)
end for
for f in createFiles do
    f.SET_ACTIVE(false)
end for
▷ Find and enable a target file
if currentPuzzle.type = DELETE then
    puzzleFile ← SelectDeleteFile()          ▷ Picks next delete file in bucket
    puzzleFile.SET_ACTIVE(true)
else
    puzzleFile ← SelectCreateFile()          ▷ Picks next create file in bucket
    puzzleFile.SET_ACTIVE(true)
    if currentPuzzle.type = MOVE then
        puzzleFile.shouldCheckContent = true
    end if
end if

introDialogue ← MakeIntroDialogue(currentPuzzle)

▷ Pick a valid level for the file
for l in levels do
    if puzzleFile.name.Contains(l.acceptableFileName) then
        options.Add(l)
    end if
end for
currentLevel ← options[random(1, options.count)]
```

Demo

Results and Discussion

Hypotheses

- Primary hypothesis - students will gain a better understanding of how to use the digital file system through playing the game.
- Expected correlations:
  - Knowledge gain and levels encountered.
  - Knowledge gain and puzzles passed.
  - Usability and participation.
  - Usability and puzzles passed.
  - Knowledge gain and usability.

Experiment

- Evaluated on two schools:
    - Private middle school in Albuquerque, NM (urban).
    - Public high school in Loving, NM (rural).
- Done over two days:
    - Day 1: Students introduced to study and given consent forms.
    - Day 2: Students took pretest, played the game, then took the posttest and completed surveys.

Activities performed on both days of evaluation.

	Activity	Duration
Day 1	Introducing project	5 minutes
	Playing tutorial	40 minutes
	Distributing and explaining data collection consent forms	5 minutes
Day 2	Reintroduce project and collect consent forms	5 minutes
	Pretest	10 minutes
	Play game	20 minutes
	Posttest	10 minutes
	Surveys and thanking participants	5 minutes

Demographics

Category Split	Demographic	Count	Percentage
Gender	Female	12	39%
	Male	17	55%
	Prefer not to say	2	6%
Computer Access	Yes	27	87%
	No	3	10%
Previous Educational Games	0-1	9	29%
	2-3	10	32%
	4-5	6	19%
	6+	5	16%
Weekly Computer Hours	0-2	2	6%
	2-5	10	32%
	5-15	10	32%
	15-30	4	13%
	30+	5	16%
Grade	Middle School	16	52%
	High School	14	45%
Region	Urban	15	48%
	Rural	16	52%
Overall	N/A	31	100%

Demographic Analysis

- Comparisons done between the different demographic groups.
  - Histograms
  - T-Tests
  - ANOVA tests
- No statistically significant differences found.
  - Closest was the t-test for educational games previously played (p-value of 0.058).
  - Suggests game is not unfair to any group.

Correlations

- Hope is that relations might indicate how the game performs.
    - Could also inform how it can be improved.
- Significance between puzzles passed and usability.
    - Students who had trouble using it had less success?
- Significance between puzzles passed and encountered.
    - Careful students timing out?

	Puzzles Encountered	Puzzles Passed	Test Improvement	Usability
Usability	-0.04/0.83	**+0.37/0.04**	+0.12/0.51	+1.0
Test Improvement	-0.00/0.96	+0.11/0.57	+1.0	
Puzzles Passed	**-0.52/0.002**	+1.0		
Puzzles Encountered	+1.0			

Split Correlations

High-scorers:

- No significant results.
  - Possibly due to smaller sample size?
- Some correlations flipped.
  - Notably different behavior.

	Puzzles Encountered	Puzzles Passed	Test Improvement	Usability
Usability	+0.24/0.45	+0.24/0.45	+0.09/0.78	+1.0
Test Improvement	+0.11/0.74	-0.27/0.40	+1.0	
Puzzles Passed	-0.40/0.20	+1.0		
Puzzles Encountered	+1.0			

Low-scorers:

- Significance between test improvement and puzzles passed.
  - Beating puzzles causing knowledge gain?
- Significance between puzzles passed and encountered.
  - Careful students timing out?

	Puzzles Encountered	Puzzles Passed	Test Improvement	Usability
Usability	-0.13/0.59	+0.43/0.06	+0.26/0.29	+1.0
Test Improvement	-0.24/0.32	**+0.58/0.01**	+1.0	
Puzzles Passed	**-0.54/0.02**	+1.0		
Puzzles Encountered	+1.0			

Knowledge Gain T-Tests

- T-tests done comparing pretest and posttest scores.
- Overall group saw benefits.

Test	Metric	Avg	SD	d.f.	T-Statistic	P-Value
Pretest (n=31)	Percentage Correct	81%	14%	30	-2.39	0.02
Posttest (n=31)	Percentage Correct	87%	12%			

Split T-Tests

- High scoring group did not benefit.
- Low scoring group did significantly.

Test	Metric	Avg	SD	d.f.	T-Statistic	P-Value
Pretest (n=12)	Percentage Correct	94%	4%	11	1.08	0.30
Posttest (n=12)	Percentage Correct	92%	9%			

Knowledge gain t-test for the high-scoring group.

Test	Metric	Avg	SD	d.f.	T-Statistic	P-Value
Pretest (n=19)	Percentage Correct	72%	11%	18	-3.56	0.002
Posttest (n=19)	Percentage Correct	84%	13%			

Knowledge gain t-test for the low-scoring group.

Comparative T-Test

- Low scoring group had more knowledge gain.

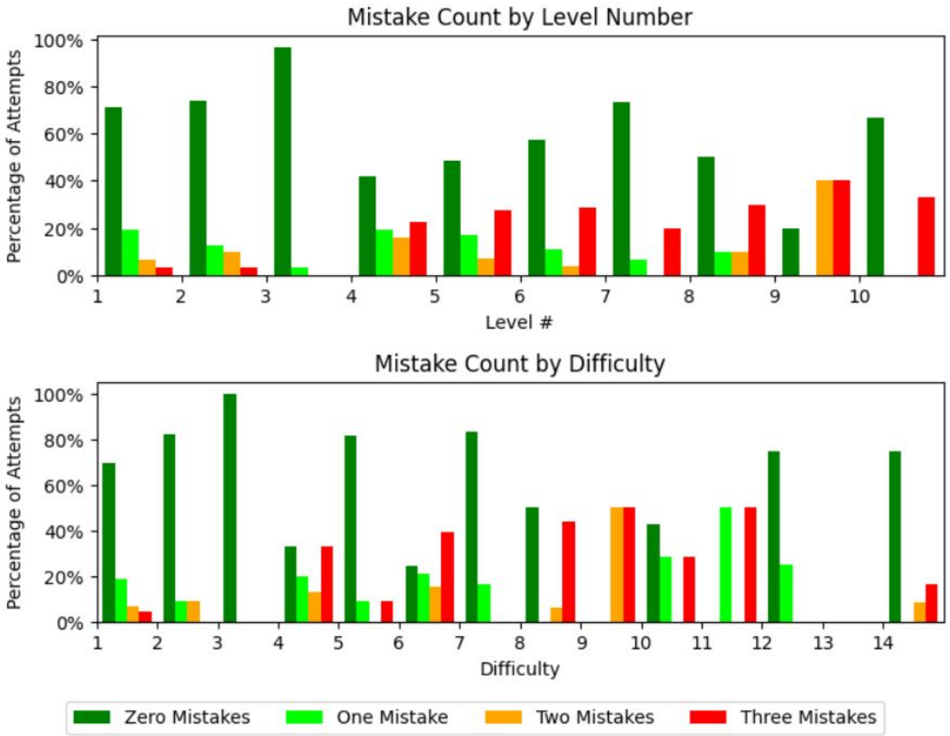
Group	Metric	Avg	SD	d.f.	T-Statistic	P-Value
High-Scoring (n=12)	Test Improvement	-3%	9%	29	-3.47	0.002
Low-Scoring (n=19)	Test Improvement	12%	15%			

T-test comparing knowledge gain between the high and low scoring groups.

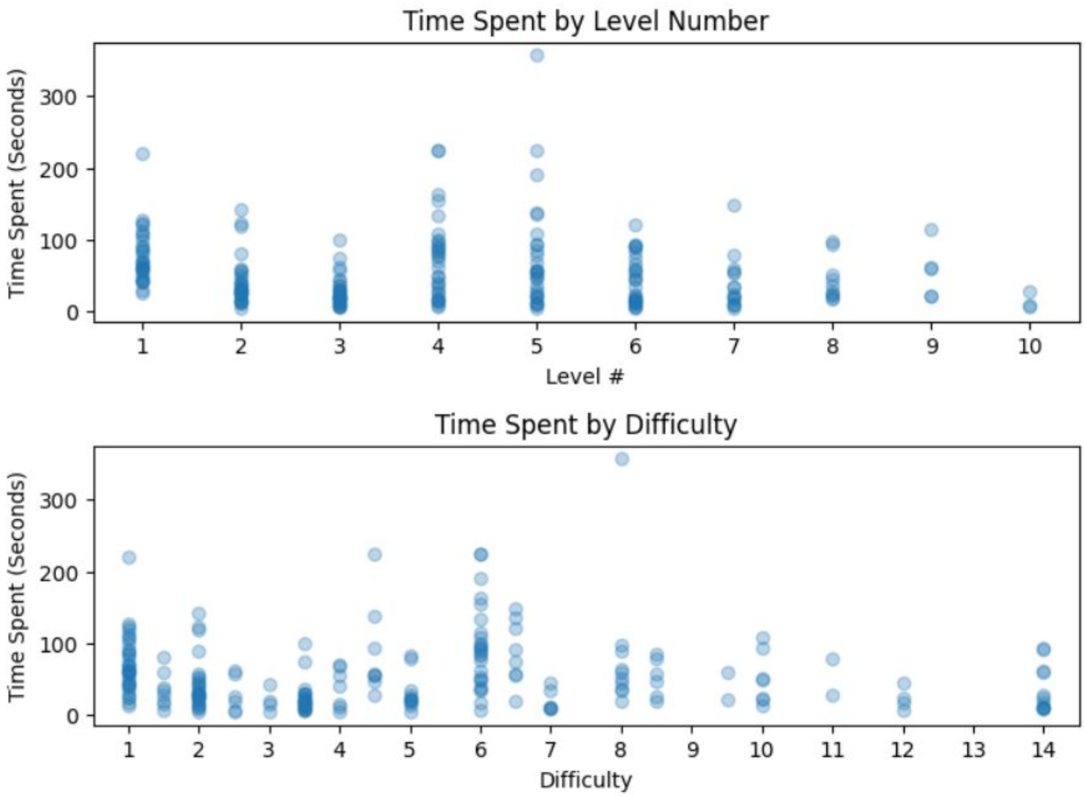
Sequence Discussion

- Mistakes/time vs. level/difficulty.
- Line plot of student paths.
- Notable difficulty transitioning from deletion to creation puzzles.
- No similar difficulty shown going from creation to movement.

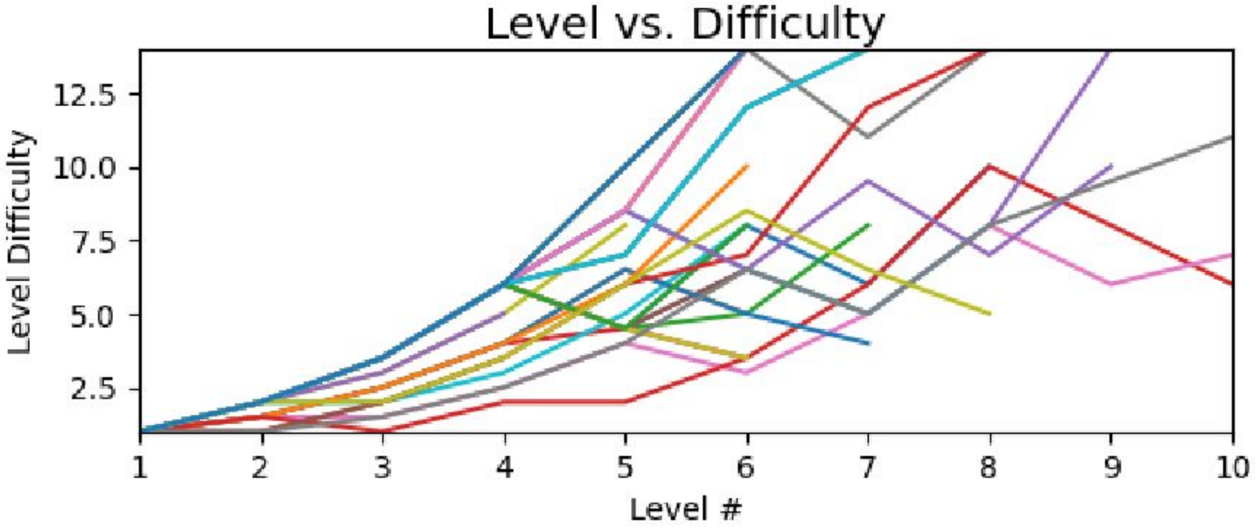
Mistake Plot



Time Plot


Time Spent by Level Number
Time Spent (Seconds)
300
200
100
0
1
2
3
4
5
6
7
8
9
10
Level #
Time Spent by Difficulty
Time Spent (Seconds)
300
200
100
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Difficulty

Level vs. Difficulty Plot

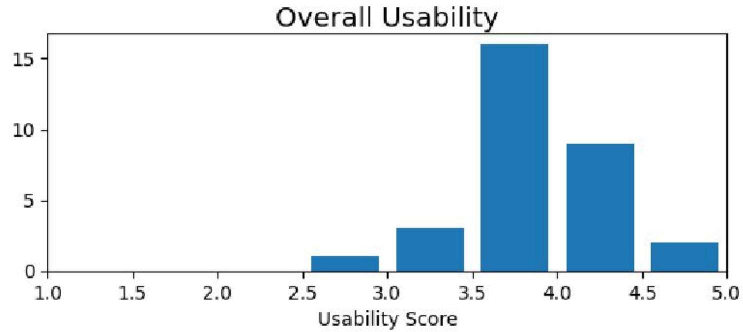


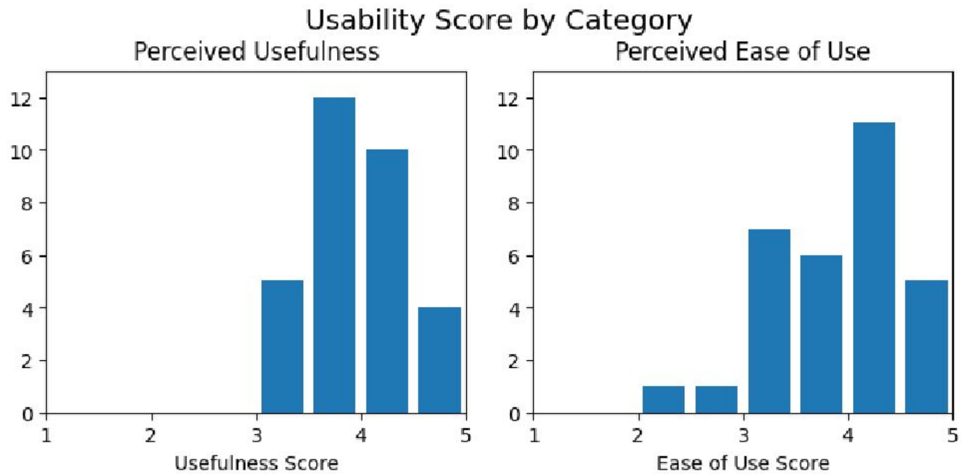
Usability Analysis

- Two points of concern: ease of use and usefulness.
    - Range between 1 (worst) and 5 (best).

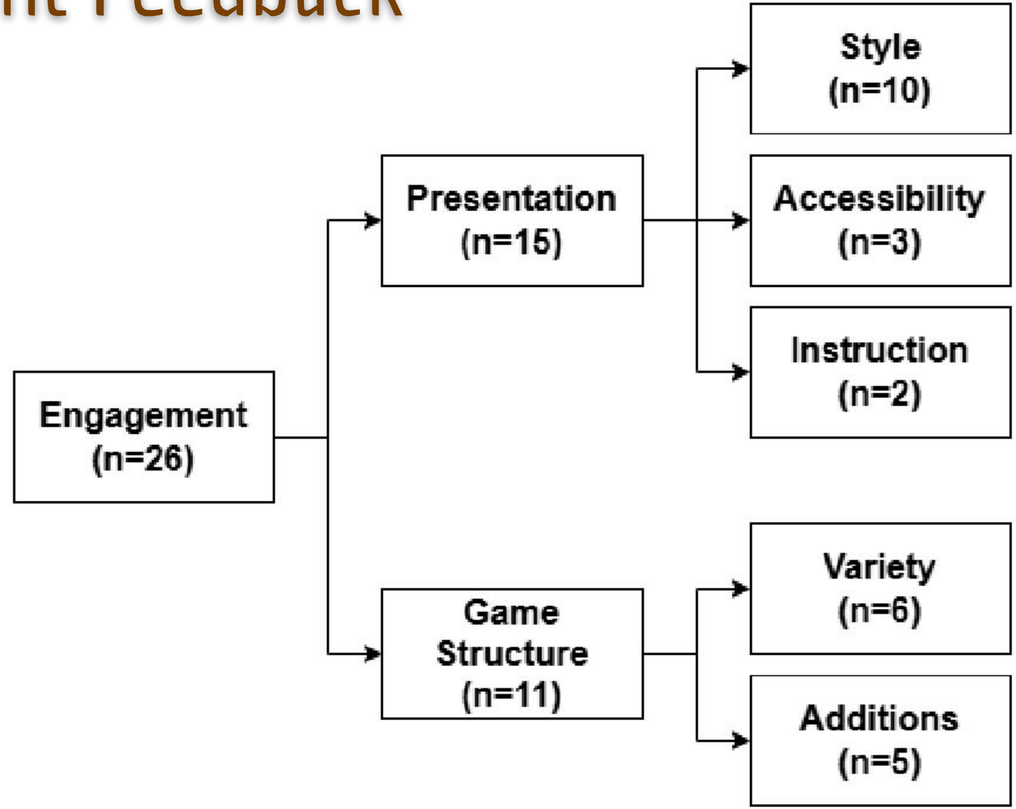
Usability Histograms

- Both metrics fairly good.
- However, some worse results for ease of use.
  - Looking at feedback, seems to be due to instructions.
- Also some notes to improve engagement, such as a character select.

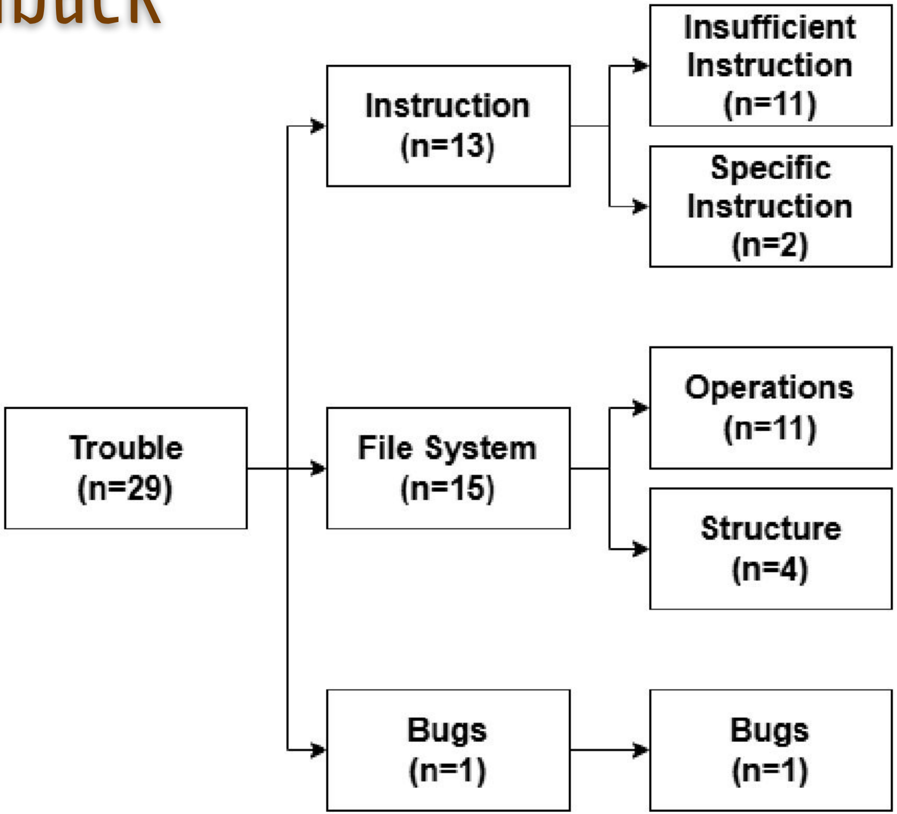




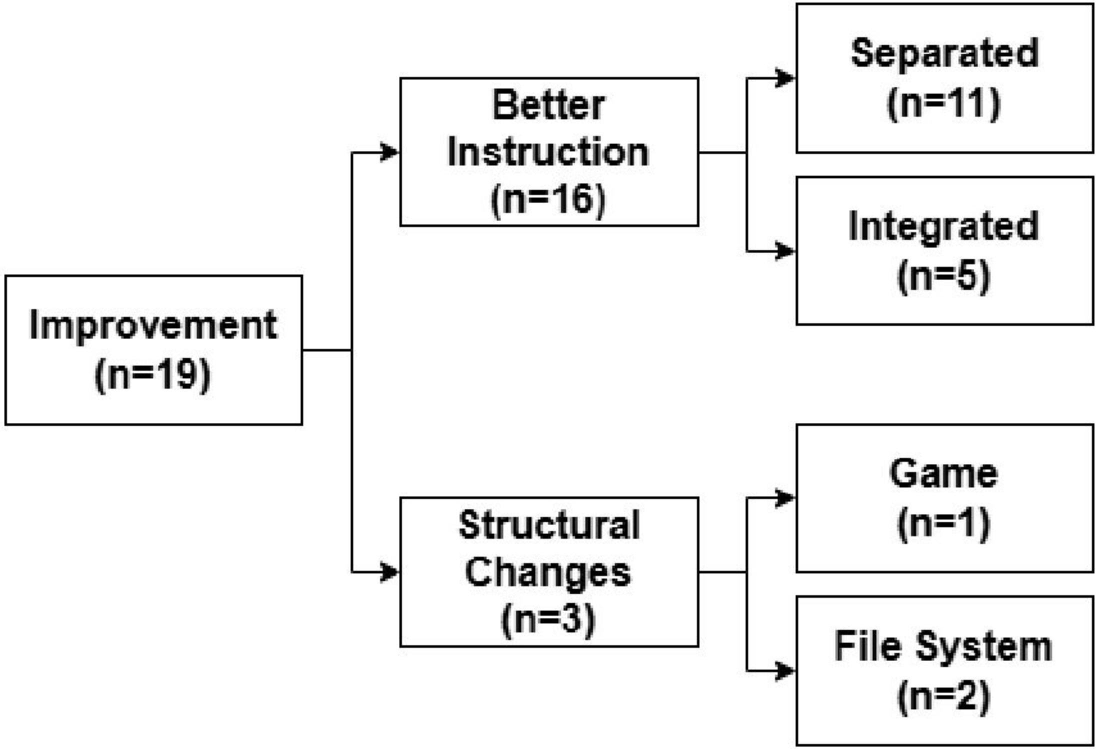
Engagement Feedback



Trouble Feedback



Improvement Feedback



Conclusion

Summary of Findings

- Overall, system was fairly successful.
    - T-tests demonstrate that it did cause knowledge gain.
- Certain drawbacks.
    - Puzzle type transitions.
    - Insufficient instruction.
- Demonstrated ability to improve student understanding via GBL, adaptive difficulty, and AIG.

Research Questions

1. How can an educational game be made to procedurally generate puzzles about the digital file system?
    a. Demonstrated by this project, can manipulate aspects of the file system to make puzzles of different difficulties.
2. How could this game determine what level should be generated, given the outcome of the previous level?
    a. Can monitor player behavior (such as mistakes made) to determine an appropriate difficulty, then generate a level based on that.
3. How does the proposed game improve student understanding of the digital file system?
    a. For those with less pre-existing knowledge, solving the puzzles provides experience on file system use and organization.

Directions for Future Work

Address shortcomings:

1. Improve puzzle type transitions.
    a. Better instruction between deletion and creation.
    b. More distinction between creation and movement.
2. Improve instruction, perhaps via integrated tutorials.
3. Improve motivation and engagement.
    a. Character select.
    b. More creative level solutions.

Expand further:

1. More focus on narrative.
2. More systems.
    a. Branching paths.
    b. Optional objectives.
3. Work on other aspects of computer/digital literacy.
    a. Scam emails.

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